

UNIT 1 · Introduction to Indian Knowledge Systems

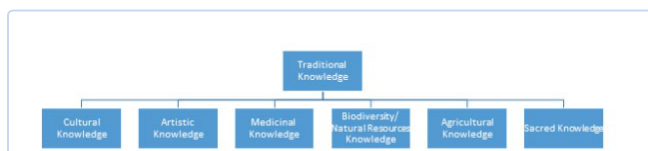
1.1 What is IKS? (Perception & prologue)

- IKS = body of knowledge and practices developed in India over **thousands of years**: philosophy, mathematics, medicine (Ayurveda), arts, architecture, agriculture, understanding of nature. Passed down through generations; promotes **harmony with nature and society**.
- Colonial education marginalised indigenous knowledge; Western epistemology dominated, creating a gap between learners and their roots. **NEP 2020** advocates reviving IKS through curriculum integration at all levels (interdisciplinary learning, indigenous languages, traditional wisdom for sustainability).
- Modern synthesis examples: traditional farming + modern tools; Ayurveda + modern healthcare.
- Languages: Sanskrit carries much of the knowledge; the motto "*Satyam vada, dharmaṃ cara*" (Taittirīya Upaniṣad; SEE MCQ). "*Sā vidyā yā vimuktaye*" = knowledge that liberates.

[BOOK p.3 ↗](#) [BOOK p.6 Importance ↗](#)

1.2 Indian Traditional Knowledge (TK): nature, scope, kinds

- TK / indigenous knowledge = knowledge, know-how, skills, practices **developed, maintained and passed from generation to generation**, often central to a community's cultural/spiritual identity.
- **Scope**: agriculture, science, technology, ecology, medicine (Ayurveda, Yoga, Unani, Siddha, Naturopathy), biodiversity, plus cultural expressions (signs, symbols).
- **How it is expressed**: mostly **oral**: tales, songs, folklore, proverbs, rituals, dance, paintings, carvings; belongs to the community as a group.
- **Kinds of TK (figure)**: Cultural, Artistic, Medicinal, Biodiversity/Natural-resources, Agricultural, Sacred knowledge. (SEE MCQ: "ecological, technical, social".)



Kinds of Traditional Knowledge (your notes)

IMPORTANT TK vs Modern science (MQP

Q1b, CIE-1)

- TK: **holistic, experiential, community-based, oral, sustainable**; knowledge = path to truth (*satya*) and harmony of self–nature–universe.
- Modern/Western: **analytical, reductionist, empirical, documented**; observation + experiment to explain/control nature.

Threats to TK & why protect it

Threats	Need to protect
• Language threat: indigenous languages (e.g. Sanskrit for Ayurveda) decline • External pressures: migration, environment, modern lifestyle, globalisation • Biopiracy: commercial exploitation of natural biochemicals/genetic material (turmeric patent by an MNC) • Colonisation, capitalism/MNCs, consumerism	• Preserve culture & identity • Protect livelihoods • Health benefits (traditional medicine for large populations) • Ecological benefits: e.g. sacred groves conserve biodiversity

1.3 Transition from traditional to Western knowledge **IMPORTANT**

- **Traditional:** Gurukula system; Nalanda, Takshashila; holistic (philosophy + science + arts + ethics); knowledge = *Vidya*, sacred, aimed at **self-realisation and societal harmony**; oral + Sanskrit texts.
- **British rule:** English medium, new curricula, modern institutions; focus shifted to **empirical, utilitarian, job-oriented** learning. Created a divide (an *epistemological shift*; MQP Q2 answer B).
- Silver lining: documentation and global exposure of Indian heritage; today we seek **synthesis** of IKS with modern science.

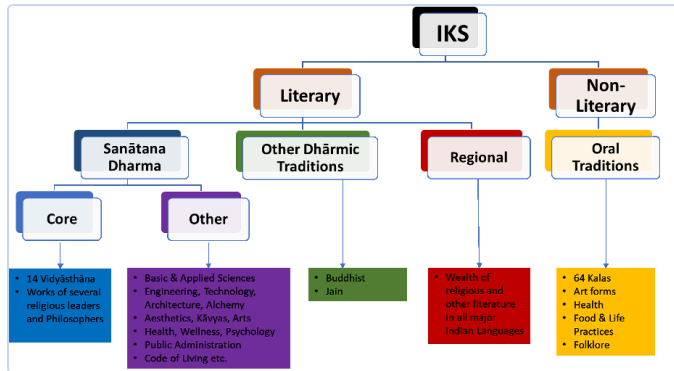
1.4 Historicity of IKS **IMPORTANT** (SEE Feb 2026 1b)

Strand	Key facts
Vedic period (c. 1500–500 BCE)	Origin; four Vedas (cosmology, spirituality, ethics); Upanishads: self (Ātman) and ultimate reality (Brahman).
Epics & classics	Rāmāyaṇa, Mahābhārata (moral guidance); Bhagavad Gītā (in Mahābhārata); Kālidāsa's <i>Śakuntalā</i> ; Patañjali's <i>Yoga Sūtras</i> (holistic well-being).
Philosophy	Six schools: Nyāya, Vaiśeṣika, Sāṃkhya, Yoga, Mīmāṃsā, Vedānta.
Mathematics	Āryabhaṭa, Brahmagupta, Bhāskarācārya: zero, decimal system, trigonometry; Brahmagupta: quadratic equations.
Astronomy	Āryabhaṭa: Earth's rotation, circumference; Varāhamihira: <i>Br̥hat Saṃhitā</i> (astronomy + meteorology).
Medicine	Ayurveda: Charaka Saṃhitā (internal medicine), Suśruta Saṃhitā (surgery).
Learning centres	Takshashila (c. 600 BCE; medicine, law, military science). Nalanda (c. 5th c. CE; library <i>Dharmagaṇja</i> ; students from China, Korea, Central Asia).
Manuscripts & scripts	Palm leaf, birch bark; Brāhmī, Kharoṣṭhī, later Devanāgarī. Architecture texts: Mānasāra, Mayamata, Samarāṅgaṇa-sūtradhāra.

BOOK p.16 Historicity ↗

BOOK p.19 Oral tradition & sūtras ↗

1.5 The IKS Corpus: classification framework **IMPORTANT**



IKS corpus tree (your notes). Memorise this shape.

- **Literary** = **Sanātana Dharma corpus** (Core: Vedas + Vedāṅgas, 14 Vidyāsthānas; Other: basic & applied sciences, engineering, architecture, alchemy, health, psychology, public administration, code of living) + **Other Dhārmik traditions** (Buddhist, Jain) + Regional literature.
- **Non-literary** = oral traditions: 64 Kalās, art forms, health practices, food, folklore.
- Buddhist: Nāgārjuna's *Rasaratnākara* (1st c. CE, alchemy). Jain: mathematics as *Gaṇitānuyoga*, *Tattvārtha Sūtra*.
- Nītiśāstra: **Rāja-nīti** (governance) and **Sāmānya-nīti** (ethics).
- CIE-1 version: Vedic corpus (Śruti) · Post-Vedic (Smṛti: Itihāsa, Purāṇas, Dharmaśāstras) · Traditional sciences · Folk & tribal knowledge.

[BOOK p.9 Corpus ↗](#)

[BOOK p.13 Caturdaśa-Vidyāsthāna ↗](#)

1.6 The Vedic Corpus **IMPORTANT**

- **Veda** = "knowledge", from root *vid* (to know). Śruti = "that which was heard": **oral tradition / revelation** (MQP Q1: B). Written form c. 1600 BCE; no definite date as oral transmission was precise.
- Four Vedas = **Chaturveda**: Ṛg, Yajur, Sāma, Atharva.

Veda	Meaning / nature	Facts to remember
R̥gveda	"Praise" · hymns; oldest	1,028 hymns, 10,552 verses, 10 Maṇḍalas ; sūktas: Puruṣa (10.90), Nāsadīya (10.129), Hiranyagarbha (10.121); 5 of 21 śākhās survive (Śākala, Bāṣkala, Āśvalāyana, Śāṅkhāyana, Māṇḍūkāyana). Aitareya Upaniṣad: Mahāvākya <i>Prajñānam Brahma</i> .
Yajurveda	Prose mantras · "book of rituals"	Two branches: Śukla (mantra and brāhmaṇa separate, "bright") and Kṛṣṇa (mixed, "dark") . 101 śākhās once, 6 now (4 Kṛṣṇa: Taittirīya, Maitrāyaṇī, Kaṭha, Kapiṣṭhala; 2 Śukla: Kāṇva, Mādhyandina).
Sāmaveda	Songs / melodies · "book of songs"	Shortest: 1,549 verses ; root of Indian classical music and dance; 3 recensions survive (Kauthuma, Jaiminīya, Rāṇāyanīya); Chāndogya and Kena Upaniṣads.
Atharvaveda	Formulas/spells for diseases and daily life	730 hymns, ~6,000 mantras, 20 books ; Muṇḍaka, Māṇḍūkya, Praśna Upaniṣads; medical/surgical verses (hymn 4.15, fracture).

IMPORTANT **Four divisions of each Veda** (SEE Feb 2026 2b, MQP 2a)

1. **Samhitā**: oldest layer: mantras, hymns, prayers, benedictions.
2. **Brāhmaṇa**: prose commentary on hymns and rituals; how to perform yajña.
3. **Āraṇyaka**: "forest texts": philosophy/symbolism behind the ritual (for hermits).
4. **Upaniṣad**: teacher–student dialogues on the self (Ātman) and Brahman; **Vedānta = end of the Vedas** (CIE-1 MCQ).

[BOOK p.28 Vedas ↗](#)

[BOOK p.30 Four Vedas ↗](#)

[BOOK p.33 Four divisions ↗](#)

1.7 Vedāṅgas: the six limbs IMPORTANT

Vedāṅga	Subject	Body part
Śikṣā	Phonetics, pronunciation, accent	Nose
Vyākaraṇa	Grammar	Mouth / face
Nirukta	Etymology, meaning of words	Ears
Chandas	Metre / prosody	Feet
Kalpa	Ritual procedure	Hands
Jyotiṣa	Astronomy / timekeeping	Eyes

- **Śikṣā:** Śikṣāvallī chapters: varṇa, svara, mātrā, bala, sāma, santāna.
- **Kalpa sūtras (4):** Śrauta (sacrifices), Grhya (domestic), Dharma (law), **Śulba** (measurement of fire altars; source of geometry).
- **Vyākaraṇa:** Pāṇini c. 400 BCE; Maheśvara Sūtras (14, from Naṭarāja's ḍamaru); Patañjali's *Mahābhāṣya*.
- **Nirukta:** Yāska's *Nighaṇṭu* (c. 600 BCE); "all names come from verbs".
- **Chandas:** Gāyatrī = 3 pādas × 8 = 24 syllables; Anuṣṭup = 4 × 8 = 32 (classical śloka).
- **Jyotiṣa:** Varāhamihira's *Pañcasiddhāntikā* (5 siddhāntas); Pañchāṅga.

IMPORTANT Vedas vs Vedāṅgas: four differences (SEE Feb 2026 1a)

	Vedas	Vedāṅgas
Purpose	Holy texts: origin of world, reason for existence	Ancillary studies to preserve, interpret, perform the Vedas
Structure	4 collections	6 limbs
Content	Hymns, incantations, rituals	Metre, grammar, linguistic analysis, astronomy
Origin	Among the oldest scriptures, c. 1500–1200 BCE	Last treatises of Vedic literature

[BOOK p.38 Vedāṅgas ↗](#)

[BOOK p.39 Śikṣā ↗](#)

[BOOK p.43 Kalpa ↗](#)

[BOOK p.45 Jyotiṣa ↗](#)

1.8 Indian philosophical systems (Darśanas) IMPORTANT

Origins in Ṛgveda hymns (Nāsadiya, Puruṣa, Vāk) → flowering in Upaniṣads. **Āstika** = accepts Vedas as valid (6 schools). **Nāstika** = denies Vedas (Cārvāka, Bauddha, Jaina).

School	Founder / text	Core idea
Nyāya	Akṣapāda Gautama · Nyāya Sūtras	Logic & epistemology ; 4 pramāṇas (perception, inference, comparison, testimony).
Vaiśeṣika	Kaṇāda · Vaiśeṣika Sūtra (6th c. BCE)	Atomism, pluralism ; physical reality made of atoms; originally 2 pramāṇas (perception, inference).
Sāṃkhya	Kapila · Sāṃkhya Kārikā	Oldest orthodox; dualism of Puruṣa (consciousness) and Prakṛti (matter) ; 3 guṇas (sattva, rajas, tamas); 25 elements.
Yoga	Patañjali · Yoga Sūtras	Sāṃkhya + God; 8 limbs (aṣṭāṅga) ; goal kaivalya.
Pūrva Mīmāṃsā	Jaimini	Interpret Vedas; ritual action (karma-kāṇḍa); salvation through Vedic action.
Vedānta (Uttara Mīmāṃsā)	Bādarāyaṇa · Brahma Sūtras	Upaniṣadic knowledge (jñāna-kāṇḍa). Advaita (Ātman = Brahman), Viśiṣṭādvaita (attributes), Dvaita (3 realities).
Cārvāka (Lokāyata)	Bṛhaspati	Only pratyakṣa valid; "extreme nāstika"; material.
Bauddha	Gautama Buddha	Non-theistic; Four Noble Truths, Noble Eightfold Path to end <i>dukkha</i> .
Jaina	24 Tīrthaṅkaras; Mahāvīra (6th c. BCE)	Anekāntavāda (many viewpoints), ahiṃsā ; only Kevalīs know full truth.

[BOOK p.56 Overview ↗](#)
[BOOK p.59 Sāṃkhya & Yoga ↗](#)
[BOOK p.63 Nyāya/Vaiśeṣika ↗](#)

[BOOK p.65 Mīmāṃsā/Vedānta ↗](#)
[BOOK p.70 Jaina ↗](#)
[BOOK p.72 Bauddha ↗](#)
[BOOK p.75 Cārvāka ↗](#)

UNIT 1 · Additions to fill gaps (continues from 1.8)

1.9 Traditional Knowledge: meaning, characteristics & scope CIE-1 3b QB Q1

- **Meaning (2M):** no single agreed definition. *Traditional* = hereditary, handed down generation to generation; *knowledge* = useful information. Developed by indigenous/local communities under their own customs and laws; now recognised by international bodies such as the UN.
- **Characteristics:** (i) passed on **orally** from generation to generation; (ii) **collectively owned** by the community, seen as a gift, not private property, so the original creator is usually unknown; (iii) learned through continuous **observation, experience and practice**; (iv) inseparable from the community's cultural life; (v) tied to local **biological resources and environment** (local, empirical, time-tested); (vi) not limited to one field of technology or the arts; use is often restricted to certain members of the community.
- **Scope in the modern context:** holistic, experiential, sustainable; applied in **medicine** (Ayurveda, Yoga), **sustainable agriculture, environmental conservation** and **eco-friendly technologies**; supports food security and livelihoods; traditional medicine is often the only affordable healthcare for poor and remote communities; gives low-energy, resource-light solutions that adapt to environmental change; may also have industrial applications.
- **Bioprospecting vs biopiracy:** bioprospecting = discovering new products from biological resources; biopiracy = using those resources/knowledge commercially without permission or fair compensation (e.g. the turmeric patent). TKDL is India's defence (Unit 5).

1.10 Kinds of TK explained: what each covers CIE-1 2d SEE 1.4 QB Q4

Kind	What it covers (use as the one-line explanation)
Cultural	Customs, festivals, folklore and social practices that define a community's identity.
Artistic	Music, dance, handloom and handicrafts.
Medicinal	Herbal and folk remedies and Ayurveda, e.g. neem for skin infection, turmeric on wounds, tulsi for cough.
Biodiversity / natural resources	Knowledge of plants, animals and plant varieties and how to use and conserve them, e.g. sacred groves.
Agricultural	Crop and seed selection, soil fertility, pest management, irrigation, harvesting, seed preservation and farm tools.
Sacred	Rituals, sacred species and sacred places, kept within spiritual practice.

IMPORTANT

Agricultural vs ecological TK (CIE-1 2d, scheme answer): **agricultural TK sustains farming systems**, whereas **ecological TK maintains environmental balance and biodiversity**.

Second classification, by domain of application (this is what the SEE MCQ "ecological, technical and social" points to; TVE is the closest match to "social"):

Class	Meaning	Examples
TEK: Traditional Ecological Knowledge	Knowledge built over hundreds or thousands of years of direct contact with the environment; location-specific; relationships between plants, animals, natural phenomena and landscape used for livelihood.	Reading the landscape and seasons for farming, fishing and forest use.
TTK: Traditional Technical Knowledge	Design and making of tools, implements and gear for natural-resource management, plus design and construction skills.	Agriculture, fisheries, forestry, handloom and handicraft; housing, water-harvesting structures, roads and bridges.
TVE: Traditional Value and Ethics	Values and ethics that grow around resource use: sacred species, forests, spaces and water bodies; seasonal restraint.	No fishing in the breeding season; harvesting forest produce only after the flowering period.

Example bank for answers (one-liners; details in Unit 4): step wells (earliest c. 550 AD), tanka and johad (Rajasthan), zabo (Nagaland), bamboo drip irrigation (North-East), vernacular housing (Kerala), region-specific ploughs, seed-storage methods, weather lore (honeybee/butterfly signs).

1.11 Nyāya perception & Patañjali's Yoga Sūtras CIE-1 2c CIE-1 3a

IMPORTANT

Two types of perception (pratyakṣa) in Nyāya (CIE-1 scheme):

- **Laukika (ordinary):** perception through the five senses in everyday experience.
- **Alaukika (extraordinary):** non-ordinary perception, of three kinds: *sāmānyalakṣaṇa* (perceiving universals), *jñānalakṣaṇa* (knowledge-based perception) and *yogaja* (through yogic power).

Nyāya also splits perception into *nirvikalpaka* (indeterminate, bare sensing) and *savikalpaka* (determinate, with naming/recognition). If a question says "two types", give laukika / alaukika as the scheme does and add this as a bonus line.

Four Yoga Sūtras to quote (CIE-1 3a, 3M for four):

Sūtra	Meaning
Yogaś citta-vṛtti-nirodhaḥ (1.2)	Yoga is the cessation of the fluctuations of the mind.
Abhyāsa-vairāgyābhyāṃ tan-nirodhaḥ (1.12)	Control of the mind comes through practice and detachment.
Tadā draṣṭuḥ svarūpe avasthānam (1.3)	Then the seer abides in his own true nature.
Sthira-sukham āsanam (2.46)	Posture should be steady and comfortable.

Eight limbs (aṣṭāṅga): yama, niyama, āsana, prāṇāyāma, pratyāhāra, dhāraṇā, dhyāna, samādhi.

1.12 Gurukula learning & Pañchakośa SEE 1.12 CIE-1 Q7

- **Mode of learning in Gurukulas:** *śravaṇa* (listening to the teacher) → *manana* (reflection and debate) → *nididhyāsana* (deep contemplation until the idea is realised). Oral recitation and memorisation carried the texts; no written examinations.
- **Pañchakośa model of education** = five layers of the self, so education develops the whole person: annamaya (physical body), prāṇamaya (vital energy), manomaya (mind), vijñānamaya (intellect), ānandamaya (bliss). Answer to CIE-1 Q7: "layers of consciousness and holistic development".

1.13 Question-bank answers: details the notes were missing QB Q6 Q7 Q8 Q11 Q20

QB	Add this detail
Q6 Mathematicians and works	Āryabhaṭa (476–550 CE), <i>Āryabhaṭīya</i> : place-value system, $\pi \approx 3.1416$, trigonometry, algebra. Brahmagupta (598–668 CE), <i>Brāhmasphuṭasiddhānta</i> : rules for zero and negative numbers, quadratic equations, astronomy and geometry. Bhāskara II (1114–1185 CE), <i>Siddhānta Śiromaṇi</i> : algebra, number systems, ideas of differential calculus centuries before Newton.
Q7 Ancient institutions	Nālandā : in present Bihar, 5th–12th c., attracted students from across Asia. Takshaśilā : in ancient Gandhāra (modern Pakistan), a centre of higher learning from about the 6th c. BCE; sometimes called the world's oldest university.
Q8 Architecture texts (any two)	Mānasāra · Mayamata · Bṛhat Saṃhitā (Varāhamihira) · Aparājita Vāstu Śāstra · Samarāṅgaṇa-sūtradhāra · Aṅka Śāstra (as listed in the QB).
Q11 Buddhist and Jain contributions	Buddhist : mathematics carried into practical sciences such as maritime activity, shipbuilding and alchemy; Nāgārjuna's <i>Rasaratnākara</i> (1st c. CE). Jain : mathematics as a core discipline (<i>Gaṇitānuyoga</i>), <i>Tattvārtha Sūtra</i> ; attempts to quantify nature, e.g. the QB cites an estimate of species as 2^{96} .
Q20 Vedānta sub-schools	Advaita (Śaṅkara): Ātman and Brahman are one. Viśiṣṭādvaita (Rāmānuja): Supreme Being (Viṣṇu) has attributes; world and souls are distinct yet inseparable modes of Brahman. Dvaita (Madhva): three separate realities: Viṣṇu, the eternal soul and matter.

TRAP

- Who gave zero?** QB Q6 credits Āryabhaṭa with zero in a positional system, while the CIE-1 scheme (5b) credits **Brahmagupta, 628 CE, with zero as an independent numeral**. For "who developed zero as a number / rules of zero" write Brahmagupta; for Q6 you can still copy the QB wording for Āryabhaṭa.
- Jain mathematician:** the QB names "Madhavacharya". The Jain mathematician usually cited is Mahāvīrācārya (*Gaṇita-sāra-saṅgraha*, 9th c.). Check with your faculty before writing either.